

Fascicule 4

II.1. Création du Réseau LAN :

 Virtual Network Editor ✕

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Bridged	Auto-bridging	-	-	-
VMnet8	NAT	NAT	Connected	Enabled	192.168.23.0
VMnet11	Host-only	-	Connected	Enabled	192.168.11.0
VMnet12	Host-only	-	Connected	Enabled	192.168.12.0

Add Network...

Remove Network

Rename Network...

VMnet Information

☐ Bridged (connect VMs directly to the external network)

Bridged to:

Automatic

Automatic Settings...

☐ NAT (shared host's IP address with VMs)

NAT Settings...

☒ Host-only (connect VMs internally in a private network)

☒ Connect a host virtual adapter to this network

Host virtual adapter name: VMware Network Adapter VMnet12

☒ Use local DHCP service to distribute IP address to VMs

DHCP Settings...

Subnet IP:

192 . 168 . 12 . 0

 Subnet mask:

255 . 255 . 255 . 0

Restore Defaults

Import...

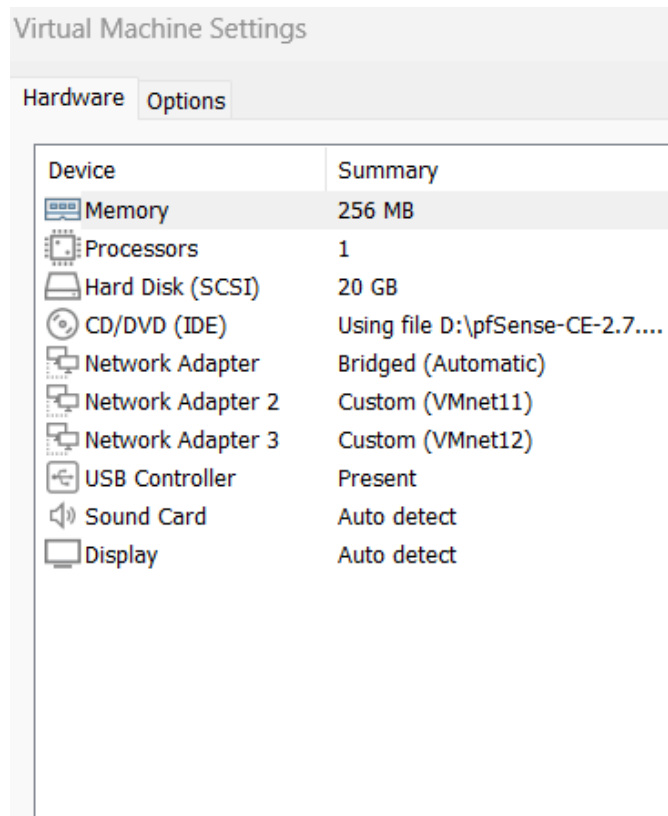
Export...

OK

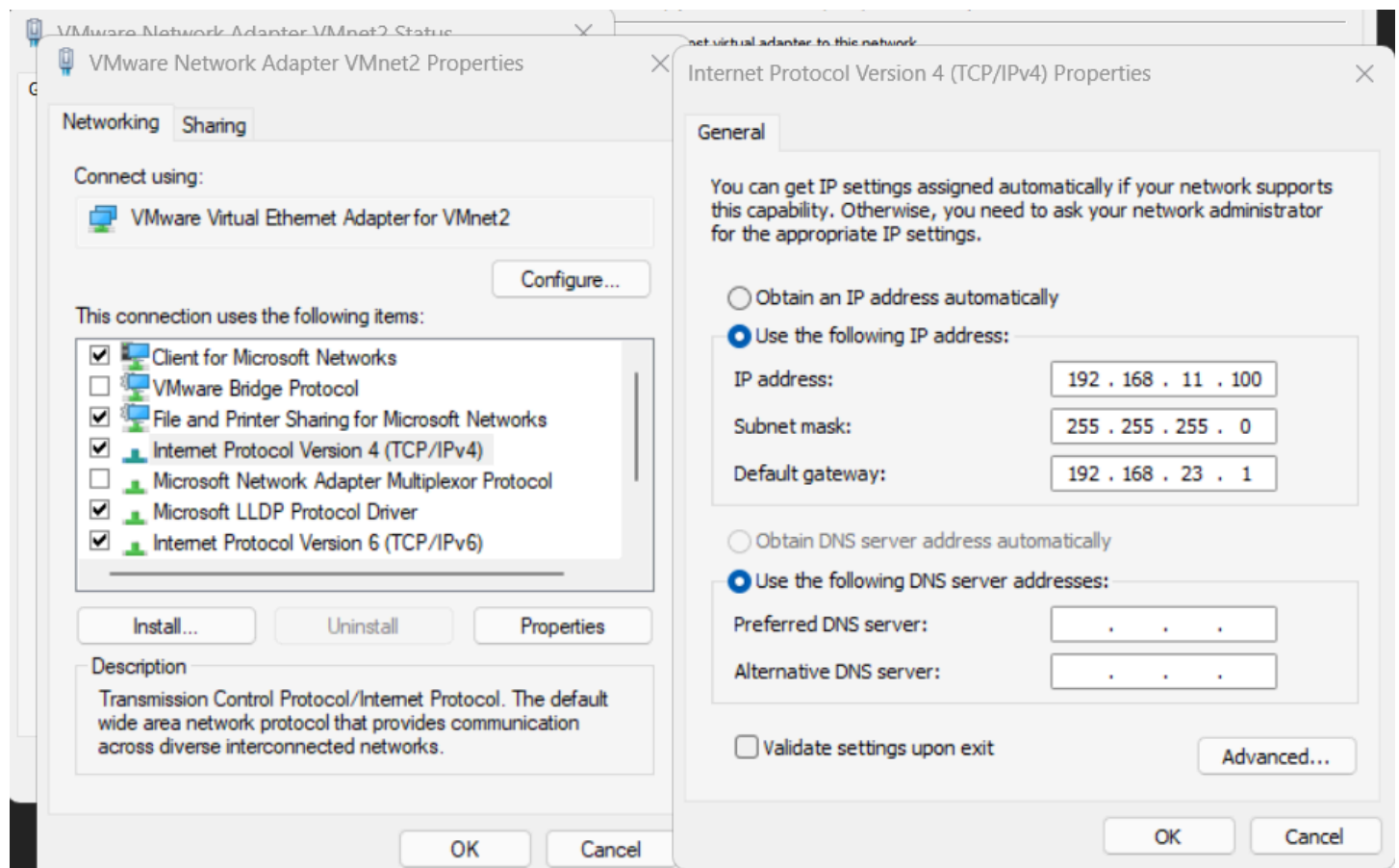
Cancel

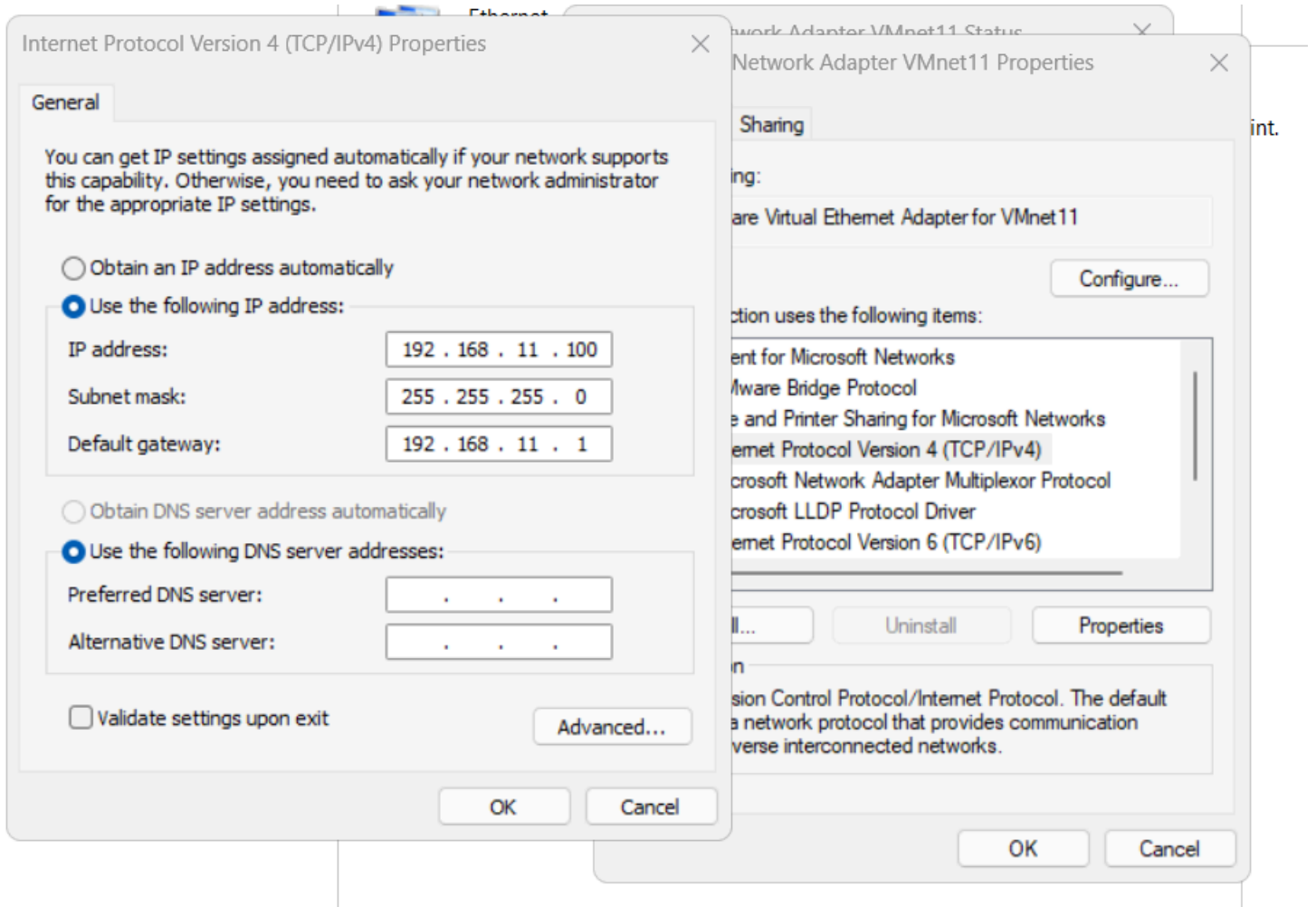
Apply

Help

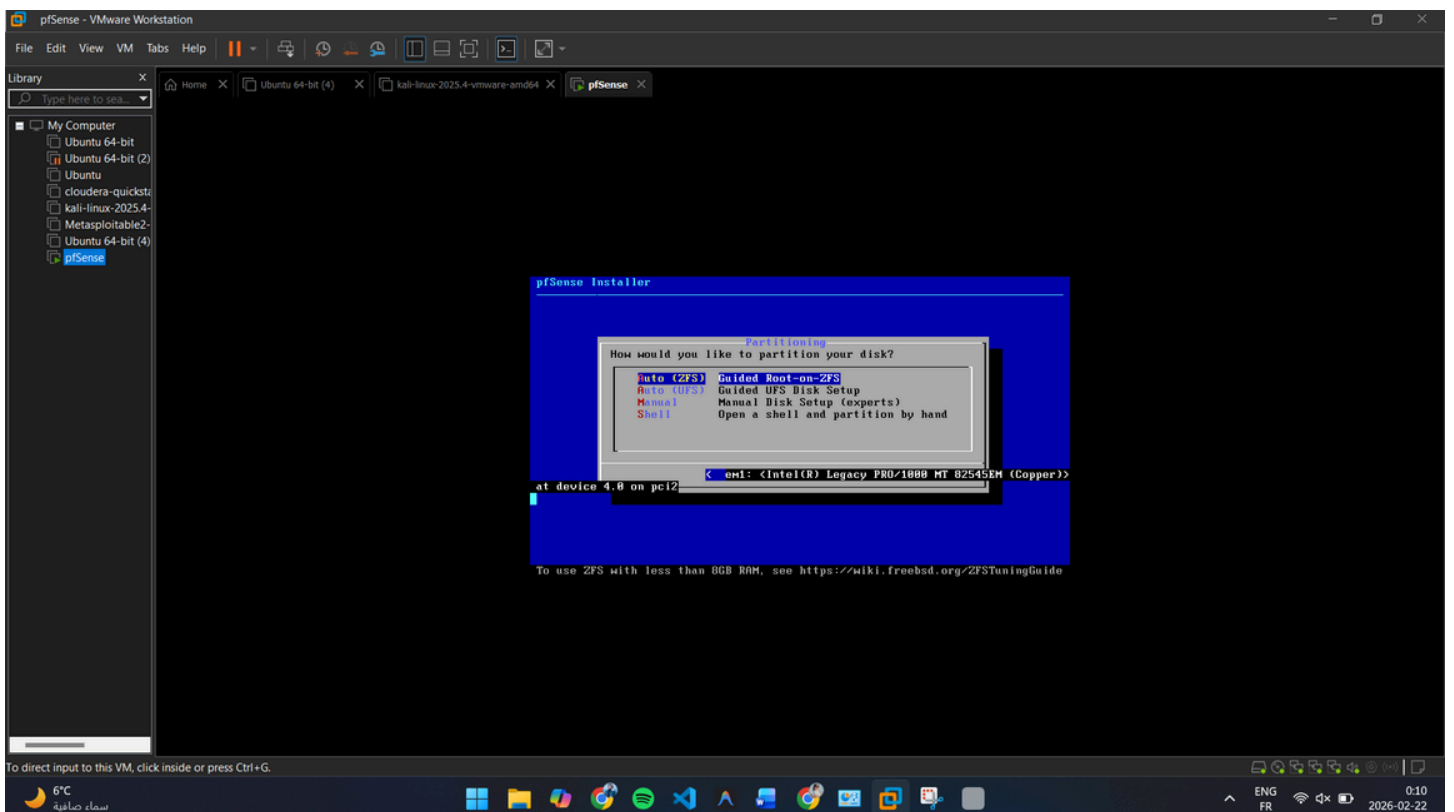


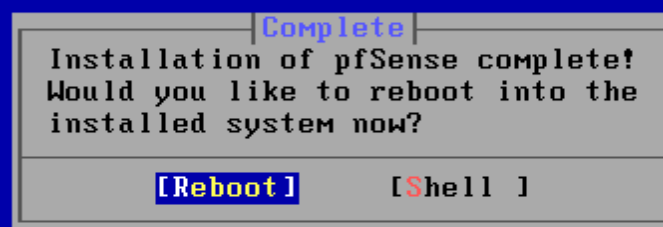
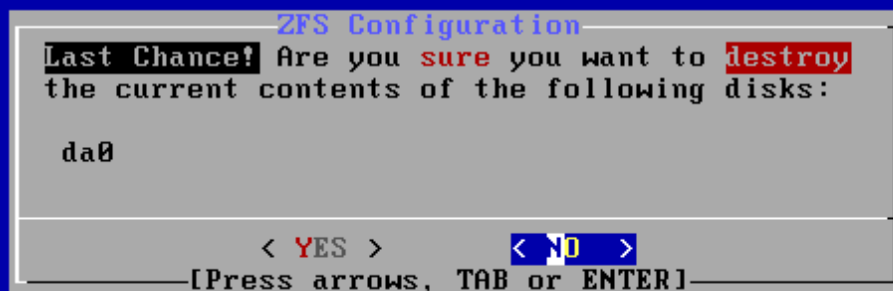
II.1. Création du Réseau DMZ :





Configuration de pfSense





Assign interfaces

say no here and use the webConfigurator to configure VLANs later, if required.

Should VLANs be set up now [y!n]? n

If the names of the interfaces are not known, auto-detection can be used instead. To use auto-detection, please disconnect all interfaces before pressing 'a' to begin the process.

Enter the WAN interface name or 'a' for auto-detection
(em0 em1 em2 or a): em0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(em1 em2 a or nothing if finished): em1

Enter the Optional 1 interface name or 'a' for auto-detection
(em2 a or nothing if finished): em2

The interfaces will be assigned as follows:

WAN -> em0
LAN -> em1
OPT1 -> em2

Do you want to proceed [y!n]? █

- 1 - WAN (em0 - dhcp6)
- 2 - LAN (em1)
- 3 - OPT1 (em2)

Enter the number of the interface you wish to configure: 1

Configure IPv4 address WAN interface via DHCP? (y/n) y

Configure IPv6 address WAN interface via DHCP6? (y/n) n

Enter the new WAN IPv6 address. Press <ENTER> for none:
> n

Enter the new WAN IPv6 address. Press <ENTER> for none:
> n

Enter the new WAN IPv6 address. Press <ENTER> for none:
>

Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) n

Please wait while the changes are saved to WAN... █

Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.11.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
255.255.0.0 = 16
255.0.0.0 = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) y
Enter the start address of the IPv4 client address range: █

Metasploitable2-
Ubuntu 64-bit (4)
pfSense

Available interfaces:

1 - WAN (em0 - dhcp)
2 - LAN (em1)
3 - OPT1 (em2)

Enter the number of the interface you wish to configure: 3

Configure IPv4 address OPT1 interface via DHCP? (y/n) n

Enter the new OPT1 IPv4 address. Press <ENTER> for none:
> 192.168.12.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
255.255.0.0 = 16
255.0.0.0 = 8

Enter the new OPT1 IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new OPT1 IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
> █

metasploitable2
Ubuntu 64-bit (4)
pfSense

The IPv4 OPT1 address has been set to 192.168.12.1/24
You can now access the webConfigurator by opening the following URL in your web browser:

<https://192.168.12.1/>

Press <ENTER> to continue.

VMware Virtual Machine - Netgate Device ID: dc241b2bb03171fb3679

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)	-> em0	->
LAN (lan)	-> em1	->
OPT1 (opt1)	-> em2	-> v4: 192.168.12.1/24
0) Logout (SSH only)		
1) Assign Interfaces		
2) Set interface(s) IP address		
3) Reset webConfigurator password		
4) Reset to factory defaults		
5) Reboot system		
6) Halt system		
7) Ping host		
8) Shell		
9) pfTop		
10) Filter Logs		
11) Restart webConfigurator		
12) PHP shell + pfSense tools		
13) Update from console		
14) Enable Secure Shell (sshd)		
15) Restore recent configuration		
16) Restart PHP-FPM		

Enter an option: █

```
valid_lft forever preferred_lft forever
eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
qlen 1000
link/ether 00:0c:29:77:60:72 brd ff:ff:ff:ff:ff:ff
inet 192.168.23.131/24 brd 192.168.23.255 scope global noprefixroute eth0
    valid_lft forever preferred_lft forever
inet6 fe80::2a0:9166:a55c:2b70/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
```

—(kali㉿kali)—[~/Desktop]

```
$ sudo ip addr add 192.168.11.100/24 dev eth0
[sudo] password for kali:
```

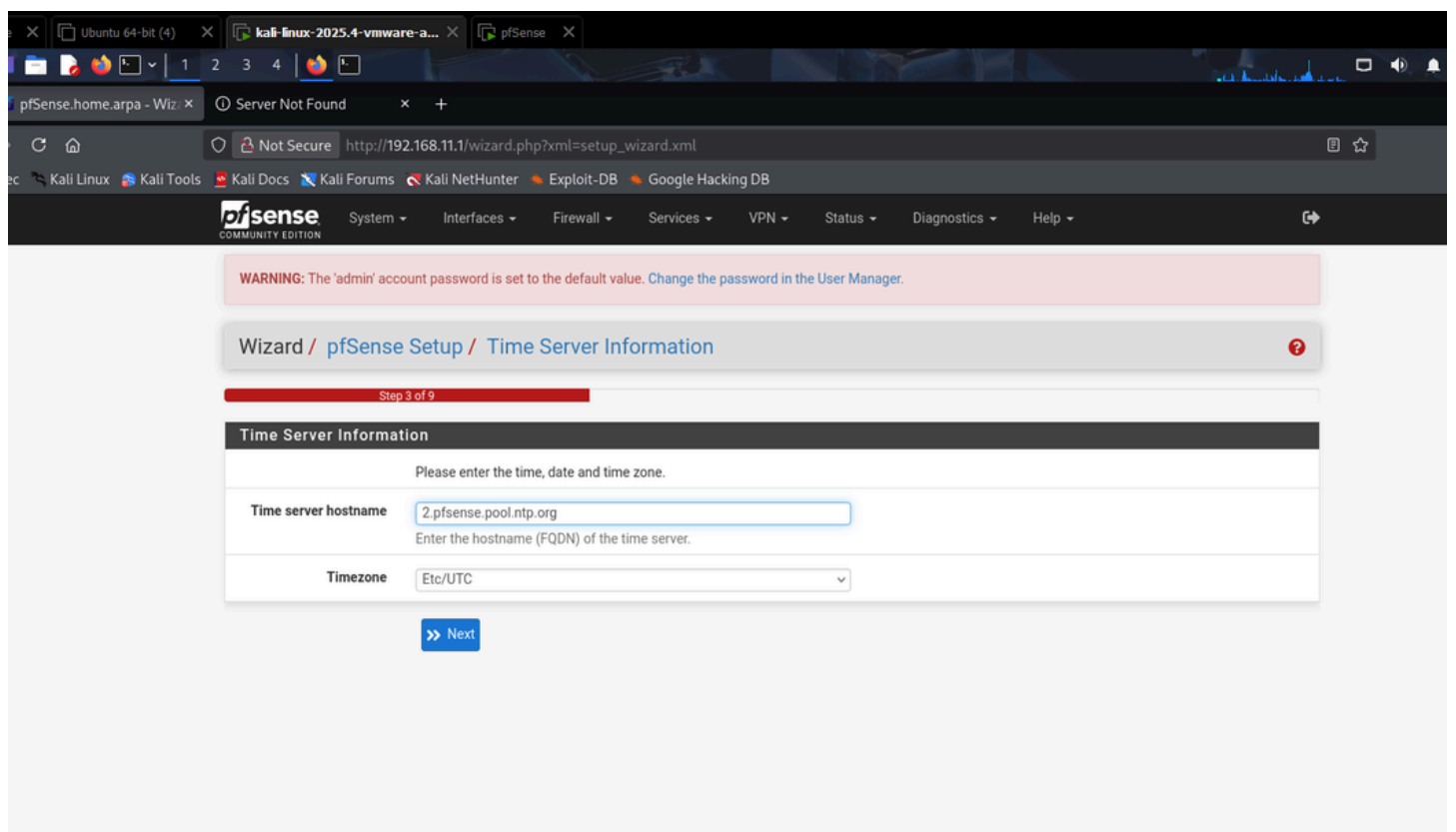
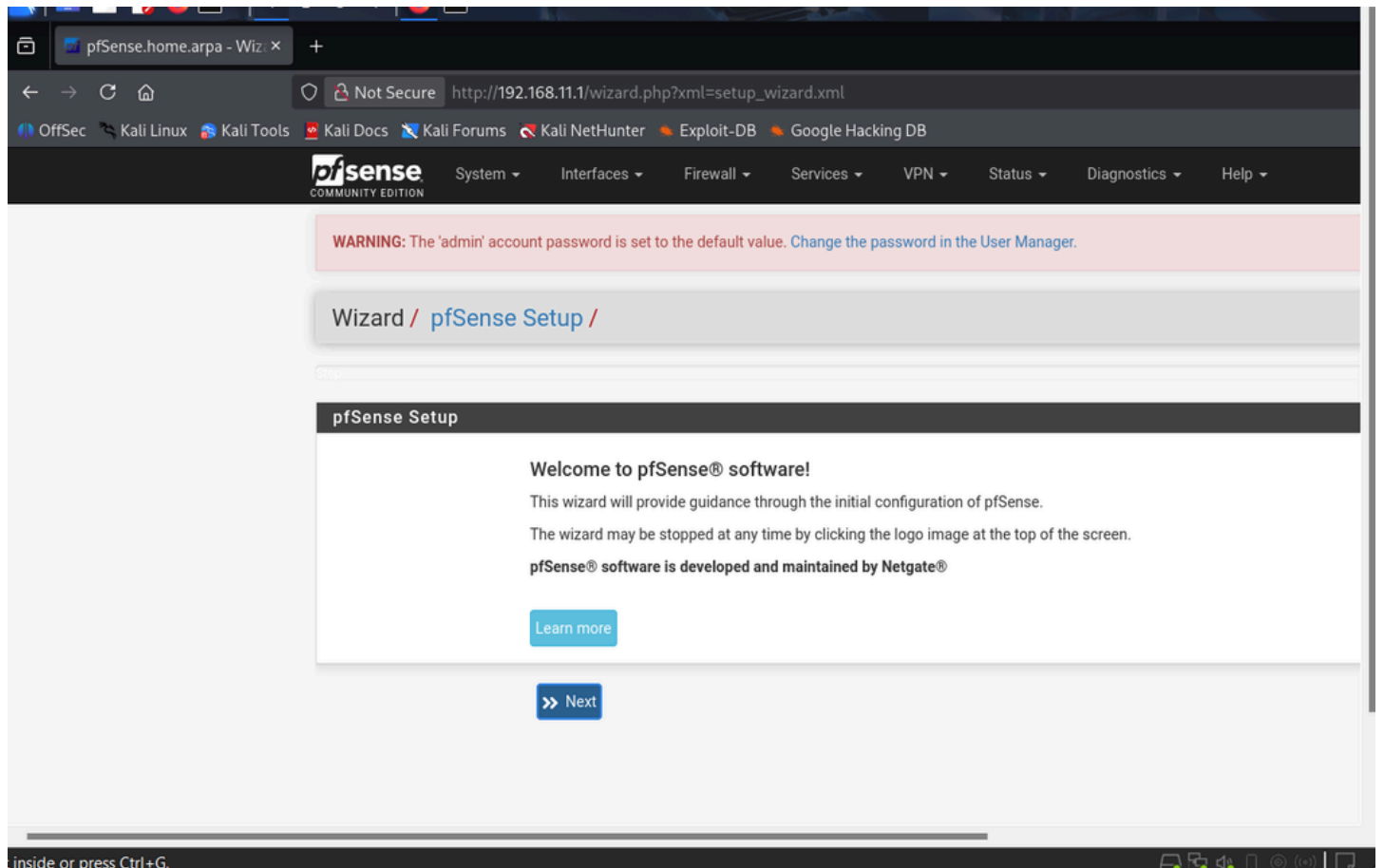
—(kali㉿kali)—[~/Desktop]

```
$ sudo ip route add default via 192.168.11.1
```

—(kali㉿kali)—[~/Desktop]

```
$ ping 192.168.11.1
```

```
PING 192.168.11.1 (192.168.11.1) 56(84) bytes of data.
+ bytes from 192.168.11.1: icmp_seq=1 ttl=64 time=0.567 ms
+ bytes from 192.168.11.1: icmp_seq=2 ttl=64 time=0.928 ms
+ bytes from 192.168.11.1: icmp_seq=3 ttl=64 time=0.729 ms
+ bytes from 192.168.11.1: icmp_seq=4 ttl=64 time=0.949 ms
+ bytes from 192.168.11.1: icmp_seq=5 ttl=64 time=0.863 ms
```



Configure WAN Interface

On this screen the Wide Area Network information will be configured.

SelectedType

General configuration

MAC Address

This field can be used to modify ("spoof") the MAC address of the WAN interface (may be required with some cable connections). Enter a MAC address in the following format: xxxxxxxxxx or leave blank.

MTU

Set the MTU of the WAN interface. If this field is left blank, an MTU of 1492 bytes for PPPoE and 1500 bytes for all other connection types will be assumed.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 (TCP/IP header size) will be in effect. If this field is left blank, an MSS of 1492 bytes for PPPoE and 1500 bytes for all other connection types will be assumed. This should match the above MTU value in most all cases.

Static IP Configuration

IP Address

Subnet Mask

Upstream Gateway

DHCP client configuration

DHCP Hostname

Wizard / pfSense Setup / Wizard completed.

Step 9 of 9

Wizard completed.

Congratulations! pfSense is now configured.

We recommend that you check to see if there are any software updates available. Keeping your software up to date is one of the most important things you can do to maintain the security of your network.

[Check for updates](#)

Remember, we're here to help.

[Click here](#) to learn about Netgate 24/7/365 support services.

User survey

Please help all the people involved in improving and expanding pfSense software by taking a moment to answer this short survey (all answers are anonymous)

[Anonymous User Survey](#)

Useful resources.

- Learn more about Netgate's product line, services, and pfSense software from our [website](#)
- To learn about Netgate appliances and other offers, [visit our store](#)
- Become part of the pfSense community. Visit our [forum](#)
- Subscribe to our [newsletter](#) for ongoing product information, software announcements and special offers.

Home x Ubuntu 64-bit (4) x kali-linux-2025.4-vmware-a... x pfSense x

pfSense.home.arpa - Sta: x Server Not Found x +

Not Secure http://192.168.11.1

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Status / Dashboard + ?

System Information ⚙️ ⌵ ⌶

Name	pfSense.home.arpa
User	admin@192.168.11.100 (Local Database)
System	VMware Virtual Machine Netgate Device ID: dc241b2bb03171fb3679
BIOS	Vendor: Phoenix Technologies LTD Version: 6.00 Release Date: Thu Nov 12 2020
Version	2.7.2-RELEASE (amd64) built on Wed Dec 6 21:10:00 CET 2023 FreeBSD 14.0-CURRENT Unable to check for updates
CPU Type	AMD Ryzen 5 5600H with Radeon Graphics AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Disabled
MDS Mitigation	Inactive
Uptime	00 Hour 15 Minutes 11 Seconds
Current date/time	Tue Feb 24 4:44:45 CET 2026

Netgate Services And Support ⌵ ⌶

Retrieving support information ⚙️

Interfaces ⚙️ ⌵ ⌶

WAN	↑	1000baseT <full-duplex>	n/a
LAN	↑	1000baseT <full-duplex>	192.168.11.1
OPT1	↑	1000baseT <full-duplex>	192.168.12.1

Not Secure http://192.168.11.1/firewall_rules.php?if=lan

Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

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Firewall / Rules / LAN

Floating WAN LAN OPT1

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	✓ 1/1.27 MiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	⚙️
☐	✓ 20/258 KiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	⚓ ⚙️ ⏏️ ⌵ ⌶ ⓧ
☐	✓ 0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	⚓ ⚙️ ⏏️ ⌵ ⌶ ⓧ

⬆️ Add ⬆️ Add 🗑️ Delete ⏏️ Toggle 📄 Copy 💾 Save ➕ Separator

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regles :

Firewall / Rules / LAN

The firewall rule configuration has been changed.
The changes must be applied for them to take effect.

✓ Apply Changes

Floating

WAN

LAN

OPT1

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	✓	2/1.30 MiB	*	*	*	LAN Address	80	*	*	Anti-Lockout Rule	
☐	✓	20/335 KiB	IPv4 *	LAN subnets	*	*	*	none		Default allow LAN to any rule	
☐	✓	0/0 B	IPv6 *	LAN subnets	*	*	*	none		Default allow LAN IPv6 to any rule	
☐	✓	0/0 B	IPv4 TCP	LAN subnets	*	*	80 - 443	*	none	Règle 1 - Navigation web	

↑ Add

↓ Add

Delete

Toggle

Copy

Save

Separator

pfSense
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System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

1

Firewall / Rules / WAN

The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor the filter reload progress.](#)

Floating

WAN

LAN

OPT1

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✗	0/1 KiB	*	RFC 1918 networks	*	*	*	*	*	Block private networks	
☐	✗	0/1 KiB	*	Reserved Not assigned by IANA	*	*	*	*	*	Block bogon networks	
☐	✓	0/0 B	IPv4 TCP	*	*	OPT1 net	80 (HTTP)	*	none		

↑ Add

↓ Add

Delete

Save

Separator

pfSense
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2

VPN / OpenVPN / Servers

Servers

Clients

Client Specific Overrides

Wizards

OpenVPN Servers

Interface	Protocol / Port	Tunnel Network	Mode / Crypto	Description	Actions
WAN	UDP4 / 1194 (TUN)	10.0.8.0/24	Mode: Remote Access (SSL/TLS + User Auth) Data Ciphers: AES-256-GCM, AES-128-GCM, CHACHA20-POLY1305, AES-256-CBC Digest: SHA256 D-H Params: 2048 bits		

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 Add

pfSense
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Firewall / Rules / LAN

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The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.

Floating WAN LAN OPT1 OpenVPN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	1 / 1.85 MiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	⚙️
☐	0 / 0 B	IPv4 *	LAN net	*	*	*	*	none			📌 ✎️ 📄 🔄 🗑️
☐	0 / 13 KiB	IPv4 *	LAN net	*	*	*	*	none		Default allow LAN to any rule	📌 ✎️ 📄 🔄 🗑️
☐	0 / 0 B	IPv6 *	LAN net	*	*	*	*	none		Default allow LAN IPv6 to any rule	📌 ✎️ 📄 🔄 🗑️
☐	0 / 0 B	IPv4 TCP	LAN net	*	OPT1 net	22 (SSH)	*	none			📌 ✎️ 📄 🔄 🗑️

⬆️ Add

⬆️ Add

🗑️ Delete

💾 Save

+ Separator

pfSense
COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

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Firewall / Rules / LAN

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Floating WAN LAN OPT1 OpenVPN

Rules (Drag to Change Order)

☒	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0 / 2.06 MiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	⚙️
☐	0 / 0 B	IPv4 *	LAN net	*	*	*	*	none		Regle1	📌 ✎️ 📄 🔄 🗑️
☐	0 / 13 KiB	IPv4 *	LAN net	*	*	*	*	none		Default allow LAN to any rule	📌 ✎️ 📄 🔄 🗑️
☐	0 / 0 B	IPv6 *	LAN net	*	*	*	*	none		Default allow LAN IPv6 to any rule	📌 ✎️ 📄 🔄 🗑️
☐	0 / 0 B	IPv4 TCP	LAN net	*	OPT1 net	22 (SSH)	*	none		Regle4	📌 ✎️ 📄 🔄 🗑️
☐	0 / 0 B	IPv4 TCP	LAN net	*	OPT1 net	22 (SSH)	*	none		Regle 5	📌 ✎️ 📄 🔄 🗑️

⬆️ Add

⬆️ Add

🗑️ Delete

💾 Save

+ Separator

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pfSense
COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

7

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Firewall / Aliases / IP

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The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.

IP Ports URLs All

Firewall Aliases IP

Name	Values	Description	Actions
SocialMedia	facebook.com, instagram.com	Regle 6	✎️ 📄 🗑️

+ Add

📶 Import

Firewall / Aliases / All



IPPortsURLsAll

Firewall Aliases All

Name	Values	Description	Actions
Ports_Teams	3478, 3479, 3480, 3481, 80, 443	Regle 7	
SocialMedia	facebook.com, instagram.com	Regle 6	

+ Add

Import

Firewall / Rules / LAN



The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.

FloatingWANLANOPT1OpenVPN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	1 / 2.25 MiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	
☐	0 / 0 B	IPv4 *	LAN net	*	*	*	*	none		Regle1	
☐	0 / 13 KiB	IPv4 *	LAN net	*	*	*	*	none		Default allow LAN to any rule	
☐	0 / 0 B	IPv6 *	LAN net	*	*	*	*	none		Default allow LAN IPv6 to any rule	
☐	0 / 0 B	IPv4 TCP	LAN net	*	OPT1 net	22 (SSH)	*	none		Regle4	
☐	0 / 0 B	IPv4 TCP	LAN net	*	OPT1 net	22 (SSH)	*	none		Regle 5	
☐	0 / 0 B	IPv4 TCP/UDP	*	*	*	Ports_Teams	*	none		Regle 7	

Add

Add

Delete

Save

Separator

Firewall / Aliases / Ports



The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.

IPPortsURLsAll

Firewall Aliases Ports

Name	Values	Description	Actions
Ports_Teams	3478, 3479, 3480, 3481, 80, 443	Regle 7	
Protocoles_Insecurises	21, 23, 80	Regle 8	

+ Add

Import



Firewall / Aliases / All



IP

Ports

URLs

All

Firewall Aliases All

Name	Values	Description	Actions
Divertissement	youtube.com, netflix.com, twitch.tv	Regle 9	
Ports_Teams	3478, 3479, 3480, 3481, 80, 443	Regle 7	
Protocoles_Insecurises	21, 23, 80	Regle 8	
SocialMedia	facebook.com, instagram.com	Regle 6	

+ Add

Import



Floating

WAN

LAN

OPT1

OpenVPN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	1 / 2.62 MIB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	
☐	0 / 0 B	IPv4 *	LAN net	*	Divertissement	*	*	none		Regle9 Heures_Travail	
☐	0 / 0 B	IPv4 *	LAN net	*	SocialMedia	*	*	none		Regle 6	
☐	0 / 0 B	IPv4 TCP	LAN net	*	*	Protocoles_Insecurises	*	none		Règle 8 : Bloquer Telnet FTP HTTP	
☐	0 / 0 B	IPv4 TCP/UDP	*	*	*	Ports_Teams	*	none		Regle 7	
☐	0 / 0 B	IPv4 TCP	LAN net	*	OPT1 net	22 (SSH)	*	none		Regle4	
☐	0 / 13 KiB	IPv4 *	LAN net	*	*	*	*	none		Default allow LAN to any rule	
☐	0 / 0 B	IPv6 *	LAN net	*	*	*	*	none		Default allow LAN IPv6 to any rule	
☐	0 / 0 B	IPv4 TCP	LAN net	*	OPT1 net	22 (SSH)	*	none		Regle 5	
☐	0 / 0 B	IPv4 *	LAN	*	*	*	*	none		Regle1	

Firewall / Rules / OPT1



The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor the filter reload progress.](#)



Floating

WAN

LAN

OPT1

OpenVPN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	0 / 0 B	IPv4 *	OPT1 net	*	LAN net	*	*	none		Règle 10 : Bloquer DMZ vers LAN	

Add

Add

Delete

Save

+ Separator



Firewall / Rules / LAN

The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor the filter reload progress.](#)

Floating
WAN
LAN
OPT1
OpenVPN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0 / 2.64 MiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	
☐	✗ 0 / 0 B	IPv4 *	LAN net	*	Divertissement	*	*	none		Règle9 Heures_Travail	
☐	✗ 0 / 0 B	IPv4 *	LAN net	*	SocialMedia	*	*	none		Règle 6	
☐	✗ 0 / 0 B	IPv4 TCP	LAN net	*	*	Protocoles_Insecurises	*	none		Règle 8 : Bloquer Telnet FTP HTTP	
☐	✓ 0 / 0 B	IPv4 *	LAN net	*	OPT1 net	*	*	none		Règle 11 : Accès employés vers DMZ	

tests :

```

(kali㉿kali)-[~/Desktop]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=110 time=30.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=110 time=32.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=110 time=32.3 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=110 time=32.1 ms
^C
— 8.8.8.8 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 30.038/31.783/32.678/1.030 ms

```